UTM FAKULTI KOMPUTERAN

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAM

SEMESTER I 2022/2023

COURSE CODE	**: SECD2523**
COURSE NAME	**: DATABASE**
YEAR / PROGRAM	**: 2/SECB, SECJ, SECV, SECR**
TIME	**:**
DATE	**:**
PLACE	**:**

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INSTRUCTION TO STUDENTS:

1. Fill in your particulars in the question-and-answer booklet.

Name	
IC No.	
Matrix No.	
Year / Program	
Course Code/ Section	
Lecturer	

2. Answer **ALL** questions in the **ANSWERING BOOKLET** given.
3. This paper consists of **THREE** parts:
   - Part A: 20 MCQ Questions
   - Part B: 5 Structured Questions
   - Part C: 1 CASE Study questions
4. Submit **BOTH question-and-answer** booklet at the end of the exam.

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PART B: STRUCTURED QUESTIONS **[60 MARKS]**

QUESTION 1 **[10 MARKS]**

a) Draw the Entity Relationship Diagram (ERD) for **EACH** of the following situations:

i. Each donor will donor one type of blood and each type of blood will be donated by at least one or more donor.

ii. A doctor will treat one or more patients and each patient will be treated by only one doctor.

iii. Each customer can rent one or more cars and each car will be rented by at most one customer.

iv. Many nurses can be monitored by a nurse.

(4 Marks)

b) The following ERDs have a few errors. Circle and describe **SIX (6)** errors below.
*(**Note:** answer in the provided attachment (Appendix 1))*

No.	ERD
i.	
ii.	
iii.	

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(6 Marks)

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QUESTION 2 **[10 MARKS]**

a) There are four (4) categories of constraints in Enhanced Entity Relationship Diagram (EERD).

Choose **TWO (2)** categories and provide **ONE (1)** example for each category.

(4 Marks)

b) A library keeps all the study materials for students and staff references. One of study material is publication. Publication can be categorized as book and article. Book cannot fall into the other category. The data stored on book includes book title, book category and book publisher. While article have article title and keyword. The publication ID uniquely identify the book and article publication information. Then the date of publication also recorded in the publication.

i) Based on the description above draw the EERD (***state participation and disjoint constraint, if any).***

ii) What is the participation and disjoint constrains for the above EERD?

(6 Marks)

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QUESTION 4	[20 MARKS]

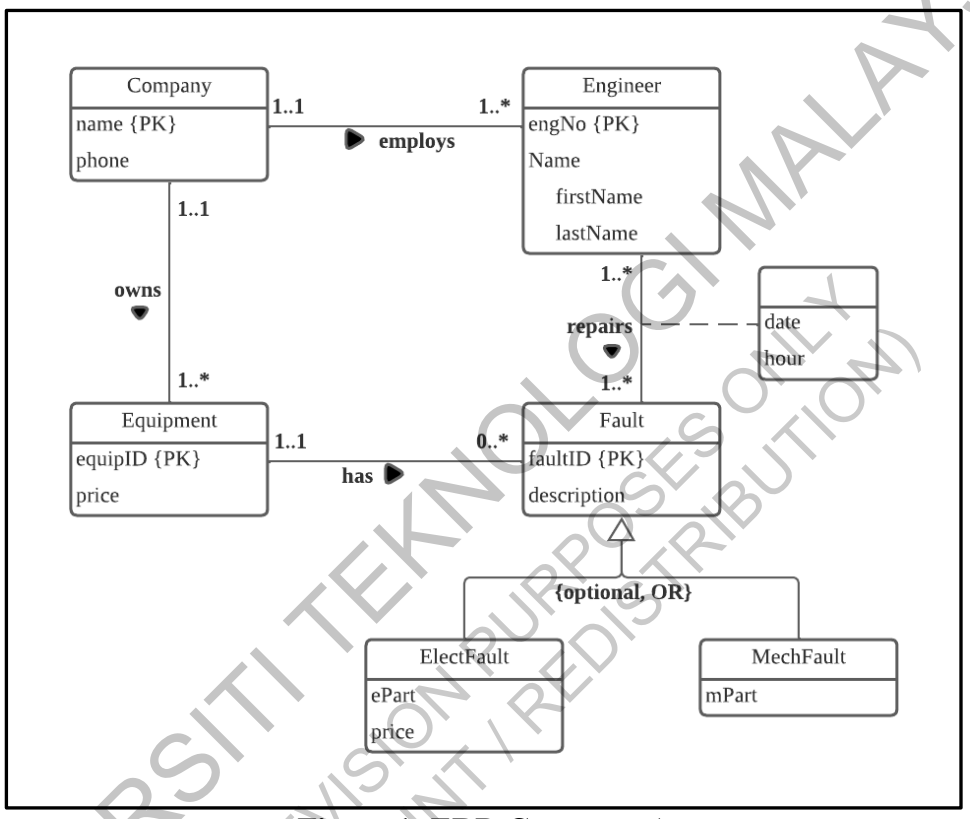


Figure 1: ERD Company A

Based on the ERD Company A given in Figure 1, answer all the following questions:

a) For relationships **owns**, **employs**, and **has**, states the type (weak or strong) of each entity that participates in the relationships. Provide your answer in the given format in Table 1.

Table 1: Format Answer

Relationship	Entity 1 - Type	Entity 2 - Type
owns		
employs		
has		

(3 Marks)

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b) Derive all relations schema from the ERD in Figure 1. State the primary key and foreign key (if any) for each schema.	
	(7 Marks)
c) Draw the logical ERD based on relations schema derived in Question 1(b).	
	(10 Marks)

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QUESTION 5 **[20 MARKS]**

Table 2: Details of patient dental appointment

staffNo	dentistName	patientNo	patientName	appointment		surgeryNo
				date	time	
S1011	Tony Smith	P100	Gillian White	12-Aug-03	10.00	S10
S1011	Tony Smith	P105	Jill Bell	13-Aug-03	12.00	S15
S1024	Helen Pearson	P108	Ian MacKay	12-Sept-03	10.00	S10
S1024	Helen Pearson	P108	Ian MacKay	14-Sept-03	10.00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Oct-03	16.30	S15
S1032	Robin Plevin	P110	John Walker	15-Oct-03	18.00	S13

a) The table shown in Table 2 is susceptible to update anomalies. Provide examples of insertion, deletion, and modification anomalies.

(8 Marks)

) Describe and illustrate the process of normalizing the table shown in Table 1 to 3NF. State any assumptions you make about the data shown in this table.

(12 Marks)

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PART C: CASE STUDY QUESTION **[20 MARKS]**

The Open University Malaysia has offered their students the newly built accommodation in the university compound. The accommodation management system is called University Hostel Management, which consist of tables Student, Lease, Hall, Flat, Room and Invoice. Please refer to the normalized tables and answer the questions below.

Student (matricNo, fName, lName, street, city, postcode, DOB, gender)

- Primary Key matricNo
- Foreign Key lNo references Lease(leaseNo)

matricNo	fName	lName	street	city	postcode	DOB	gender	lNo
MC454693	Azrul	Hakim	Jalan Indah 2	Kuala Lumpur	56100	21 June 1987	Male	L01
MC129048	Maimon	Naim	Jalan Paya 81	Batu Pahat	45000	30 May 1981	Female	L02
PC284980	Nabil	Zain	Jalan Cemerlang 1	Ayer Keroh	32900	6 April 1990	Male	L03
MC683469	Adi	Bahrin	Jalan Menteri 3	Johor Bahru	88100	2 August 1988	Male	L04
UC752371	Zatil	Ahmad	Jalan Indah 4	Kuala Lumpur	56100	16 November 1982	Female	L05

Lease (leaseNo, duration, matricNo, placeNo, dateEnter, dateLeave)

- Primary Key leaseNo
- Foreign Key mNo references Student(matricNo)
- Foreign Key pNo references Room(placeNo)

leaseNo	duration	mNo	pNO	dateEnter	dateLeave
L01	6months	MC454693	P01	1-01-22	30-6-22
L02	6months	MC129048	P02	1-01-22	30-6-22
L03	12months	PC284980	P03	15-01-22	14-01-23
L04	8months	MC683469	P04	15-01-22	14-08-22
L05	3months	UC752371	P05	1-06-22	31-08-22

Hall (halINo, hName, hAddress, hTeINo)

- Primary Key halINo

hallNo	hName	hAddress	hTelNo
h0101	Hall Cendana	Jalan Rahman	07-89294000
h0102	Hall Sapura	Jalan Razak	07-89295000
h0103	Hall Cempaka	Jalan Abdullah	07-89296000

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Flat (flatNo, fAddress, noOfRooms)

- Primary Key flatNo

FlatNo	fAddress	noOfRooms
F001	Flat Merpati	3
F002	Flat Pipit	4
F003	Flat Helang	2

Room (placeNo, roomNo, monthlyRent, flatNo, hallNo)

- Primary Key placeNo
- Foreign Key halINo references Hall(hallNo)
- Foreign Key flatNo references Flat(flatNo)

placeNo	roomNo	monthlyRent	fNo	hNo
P01	01-23	500	F001	h0101
P02	02-56	60	F002	h0102
P03	03-11	60	F003	h0103
P04	04-24	700	F002	h0102
P05	06-35	300	F003	h0103

Invoice (invoiceNo, semester, paymentDue, datePaid, paymentMethod, leaseNo)

- Primary Key invoiceNo
- Foreign Key lNo references Lease(leaseNo)

invoiceNo	semester	paymentDue	datePaid	paymentMethod	lNo
PO79900	3	30-Sept-2023	15-Sept-2023	Cash	L01
PO35850	2	30-Sept-2023	1-Sept-2023	Online transfer	L02
PO84909	1	1-June-2023	2-June-2023	Card credit	L03
PO09786	2	30-Oct-2023	29-Oct-2023	Online transfer	L04
PO86452	2	30-Sept-2023	30-Sept-2023	Debit card	L05

Write/Construct the SQL statements for each query in questions 1 to 9:

1. A list of all hall number, hall name, hall address and their phone number.

(2 Marks)

2. For Flat Merpati from table Flat, retrieve the flat number, flat address and number of rooms.

(2 Marks)

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3. For each of female students that have been applying for an accommodation in university, list their first name, last name, matric number, birth date and home city.

(3 Marks)

4. Display the invoice number, date paid and payment due for second semester students who have made the accommodation payment using online transfer. Use Invoice Number, Date Paid and Payment Due from table Invoice as the column aliases.

(2 Marks)

5. Display the student first name and their gender alphabetically in order of student name. Use Student Name and Student Gender an alias for your column headings.

(2 Marks)

6. Display all the information about student information and their invoice.

(2 Marks)

7. A list of room number which has the floor number in their hall number and their rental rate. Sort the list by their rental rate in ascending order.

(2 Marks)

8. Use an ON clause to join the student and lease table so that you display the matric number, student first name, student last name, city, postcode, DOB, gender, lease number, leaseNo, duration, place number, date enter the place and date leaving the place. However, show the results for the student that rent the room form 6 months only

   Find the error in the SQL statement below. Then, write the correct SQL statement.

   ```
   SELECT matricNo, fName, lName, city, postcode, DOB, gender, lNo
   FROM student JOIN lease
   ON (lNo = lease_no)
   WHERE monthlyrent = 60;
   ```

(3 Marks)

9. Produce the result of the following SQL statement:

   ```
   SELECT invoiceNo, semester and paymentMethod
   FROM Invoice;
   ```

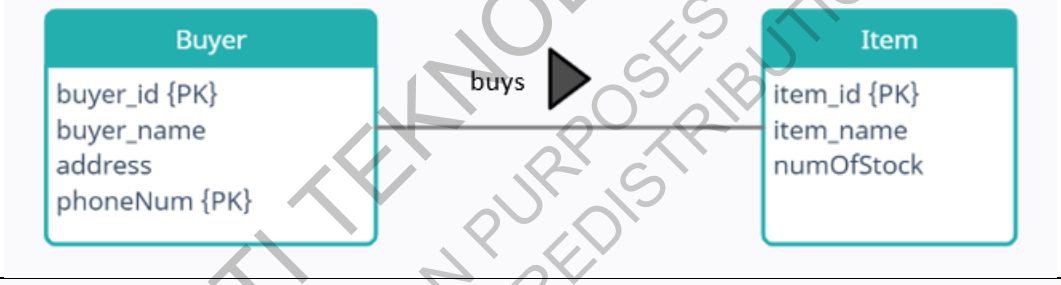
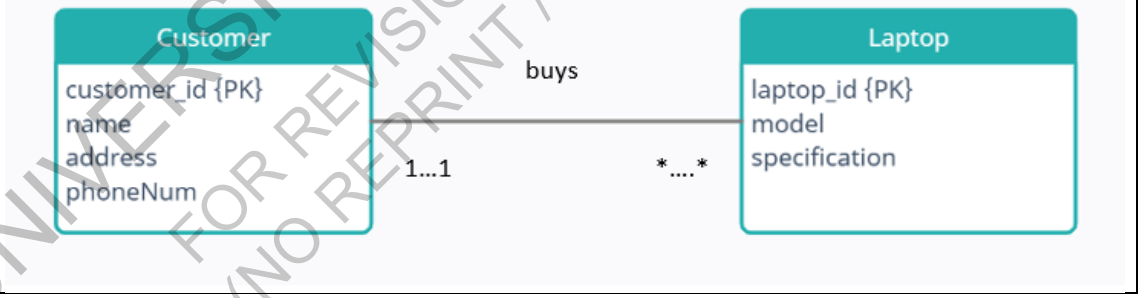
(2 Marks)

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APPENDIX 1

Circle the error and describe **SIX (6)** errors below.

No.	ERD
i.	
ii.	
iii.	

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Description on errors: